Comparing microbial growth models and temperature effects on fitted parameters

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Abstract

Microbial growth curves differ in their observed growth phases and measurement scales, which can affect model comparison and parameter interpretation. I compared exponential, logistic and Gompertz models using nonlinear least squares on a compiled dataset containing 4,387 observations and 305 curves from ten studies. A common screening rule retained 290 curves; 284 had complete three-model comparisons. Logistic had the lowest corrected Akaike information criterion (AICc) in 123 curves, Gompertz in 92 and exponential in 69. Removing comparisons with numerical warnings retained the same overall ranking. Model preference varied across supplied species labels, but species and study effects were not separated statistically. A separate temperature analysis used logistic parameters from comparable optical-density curves in one study. Strict screening retained 41 of 88 curves and eight series with at least three temperatures. All eight had positive descriptive slopes of log growth rate against temperature, although individual series were not always monotonic. Plateau estimates showed less consistent temperature patterns. Bounded models were preferred for most curves, but could not follow growth followed by decline. The temperature results describe selected experimental series and OD-based parameters rather than a general response across the compiled dataset.

1 Introduction

Microbial growth depends on both the ability of cells to reproduce and the conditions in which they grow. A culture may initially increase rapidly, then slow as resources become limiting. Growth models summarise these changes through parameters such as growth rate and plateau height. However, an experiment that ends during rapid growth provides different information from one that includes a plateau or subsequent decline.

The exponential model assumes a constant per-capita growth rate. The logistic model represents a rate that decreases as the population approaches a limit, while the Gompertz function allows an asymmetric sigmoid. These alternatives have been compared for bacterial growth curves (Zwietering et al., 1990). I expected logistic or Gompertz to receive greater support when observations covered both growth and a plateau. Exponential could be sufficient where only the increasing phase was observed. Comparing models with a penalty for additional parameters helps assess these alternatives without assuming that the most flexible model is always preferable (Johnson and Omland, 2004).

Temperature can affect growth rate and the final population measurement differently. Bacterial growth rates often increase with temperature below their optimum (Ratkowsky et al., 1982). In contrast, plateau height also depends on resource use, cell size and the measurement method; Bernhardt et al. (2018) linked temperature-dependent carrying capacity to metabolism and body size. I therefore expected increasing fitted rates within suitable temperature ranges, but no common direction for plateau estimates across organisms.

Here I ask two questions: which growth models receive most support across the compiled curves and organism labels, and how do logistic rate and plateau estimates vary with temperature within comparable series? I first compare three models across the dataset. I then restrict the temperature analysis to one study with a common medium and measurement method, so that differences between studies are less likely to obscure the temperature patterns.

2 Methods

2.1 Data preparation

The supplied growth and metadata files contained 4,387 observations from ten cited studies (Table 1). Curves were defined by citation, species label, temperature, medium, replicate label and both measurement and time units. This produced 305 curves. Measurement labels included optical density at 595 nm, N, CFU and dry weight. Labels and observations were retained as supplied; no conversion to cell counts or common biomass units was attempted.

Observations were sorted within curves and time was expressed relative to the first observation. Negative original times, nonpositive measurements and repeated times were flagged for review, without deleting or averaging observations for fitting. Only curves with at least six distinct observation times and a positive time span were fitted. This analysis choice excluded 15 curves and retained 290. It provided a common candidate set and avoided fitting three-parameter curves to extremely small datasets, but did not guarantee adequate coverage of growth phases.

Table 1: Study composition and fitting coverage. Counts describe curves, not independent species. Measurement units are the supplied labels. "Complete" means all three models converged and had finite comparison criteria.

Study	Original	Screened	Complete	Units
Bae et al. (2014)	88	88	88	OD595
Bernhardt et al. (2018)	25	25	25	N
Galarz et al. (2016)	72	72	72	CFU
Gill and DeLacy (1991)	32	29	25	N
Phillips and Griffiths (1987)	18	9	9	N
Roth and Wheaton (1962)	35	34	33	CFU
Silva et al. (2018)	17	17	17	N
Sivonen (1990)	8	8	8	Dry weight
Stannard et al. (1985)	6	4	3	CFU
Zwietering et al. (1994)	4	4	4	N

2.2 Models and numerical fitting

For time t since the first observation, the fitted functions were

$$y_{\exp}(t) = N_0 \exp(rt), \tag{1}$$

$$y_{\log}(t) = \frac{K}{1 + [(K - N_0)/N_0] \exp(-rt)}, \tag{2}$$

$$y_{\mathrm{Gomp}}(t) = K \exp[-b \exp(-gt)]. \tag{3}$$

In the exponential and logistic models, N_0 is the fitted initial measurement and r is a rate constant in hours^{-1}. In the bounded models, K is the fitted plateau. Gompertz b controls the curve position and g is its rate constant; its initial value is $K \exp(-b)$. The logistic function solves the density-dependent equation $dy/dt = ry(1 - y/K)$. Gompertz g and logistic r have different meanings and were not combined in the temperature analysis. All fitted parameters were positive. Logistic can decline when $N_0 > K$, whereas this Gompertz form is increasing.

Parameters minimised the sum of squared residuals on each curve's original measurement scale. To improve numerical behaviour, time was divided by duration and measurements by their maximum absolute value during optimisation. Parameters were converted back to original units afterwards. Bounded SciPy least-squares optimisation used three starting values for exponential, six for logistic and twelve for Gompertz. The converged solution with lowest residual sum of squares was retained. Initially, each start allowed 2,000 function evaluations; failed model–curve pairs alone were retried with 10,000. Successful original fits were not replaced by retries.

On the scaled axes, exponential started at the mean measurement with rates 0.1, 1 and 3. Logistic used the first measurement with rates 1, 5 and 20 and capacities 1 and 2. Gompertz combined capacities 1 and 2, shape parameters 1 and 4, and rates 1, 5 and 20. Initial measurement guesses were bounded below by 0.01; optimisation bounds were 10^{-10} to infinity for all parameters.

Numerical review flags identified a rank-deficient fitted Jacobian, a non-finite condition number or one above 10^8, or a scaled parameter at or below 10^{-8}. These thresholds are diagnostics of possible instability, not formal tests of biological identifiability. Predicted curves and residuals were also inspected, with attention to decline, weak growth and missing plateaus (Figure 1).

2.3 Model comparison

For comparison, residuals were provisionally treated as independent Gaussian errors with constant variance within each curve. With n observations, residual sum of squares RSS and fitted variance RSS/n, the maximised log likelihood was

$$\widehat{\ell} = -\frac{n}{2}\left[\log(2\pi) + 1 + \log(\mathrm{RSS}/n)\right]. \tag{4}$$

The parameter count k included error variance: three for exponential and four for logistic and Gompertz. I calculated $\mathrm{AIC} = -2\widehat{\ell} + 2k$ (Akaike, 1974), the finite-sample adjustment

$$\mathrm{AICc} = \mathrm{AIC} + \frac{2k(k+1)}{n-k-1}, \tag{5}$$

and $\mathrm{BIC} = -2\widehat{\ell} + k\log n$ (Schwarz, 1978). The AICc adjustment follows the small-sample model-selection approach of Hurvich and Tsai (1989); its application here is an approximation for nonlinear fits under the stated error model.

AICc was the primary criterion and BIC a sensitivity comparison. Comparisons required three converged fits and finite criteria on the same observations. Differences were calculated from each curve's minimum; raw criteria and RMSE were not averaged across different measurement units. A unique minimum used an absolute numerical tie tolerance of 10^{-8}. Models within two AICc units were recorded as having similar relative support. A second summary removed any curve with a numerical flag in any candidate model, retaining a common three-model comparison set.

2.4 Temperature analysis

Temperature analysis used the 88 Bae-study curves labelled TSB and OD595 (Bae et al., 2014). One model form, logistic, was retained at every temperature to keep parameter definitions con-

sistent. Series preserved exact citation, species or strain label, medium, replicate label and measurement unit. A replicate label shared across temperatures was not treated as proof of a paired design.

Strict screening required a converged logistic fit without a numerical flag, $K > N_0$, an inflection within the observed period, an observed measurement range greater than five RMSE, and a final prediction at least 90% of K. At least three distinct times had to fall between 10% and 90% of fitted K. A late-decline diagnostic compared the median of the three highest time-averaged measurements with the median of the last three, relative to the increase from the first measurement to that high value; declines above 20% were excluded. Repeated times were averaged only for this diagnostic.

A relaxed sensitivity screen used three RMSE, 80% of K, two transition times and a 30% decline limit. These thresholds were analysis choices used to select curves with visible growth and plateau phases. Series with at least three retained temperatures were plotted. Straight-line slopes of $\log r$ against temperature summarised direction across the observed range. They were descriptive summaries, rather than estimates of an optimum temperature. Logistic AICc support was displayed but was not an additional selection rule.

2.5 Computing tools

I used Python 3.14.5 throughout the analysis to keep data processing and fitting in one environment. pandas 3.0.6 handled the grouped data and summary tables, NumPy 2.5.4 provided array operations, and SciPy 1.18.1 supplied nonlinear least-squares optimisation (Virtanen et al., 2020). Matplotlib 3.11.2 produced vector PDF figures. I ran the scripts from Windows PowerShell within a virtual environment; separate R or bash analyses were unnecessary. LaTeX formatted the report, and BibTeX managed author–year citations.

3 Results

3.1 Fitting coverage and overall model preference

After retries, exponential and logistic converged for all 290 screened curves, while Gompertz converged for 284. Six curves therefore lacked a complete comparison. Of the 284 complete comparisons, 50 had at least one numerical flag and 234 had none. Retry convergence did not by itself remove numerical concerns.

Logistic had the unique minimum AICc in 123 of 284 curves (43.3%), Gompertz in 92 (32.4%) and exponential in 69 (24.3%; Figure 2). Logistic and Gompertz were each within two AICc units of the minimum in 172 curves; exponential was within two units in 76. These counts overlap because more than one model can be within two units for the same curve.

After removing all numerically flagged comparisons, the AICc counts were 106 logistic, 80 Gompertz and 48 exponential among 234 curves. The ranking was unchanged. BIC selected logistic in 148, Gompertz in 117 and exponential in 19 of the 284 full comparisons. In the subset without numerical flags, the corresponding BIC counts were 129, 99 and six.

3.2 Variation across organisms and curve shapes

Complete comparisons covered 45 supplied species, strain or community labels. All 12 curves from the three *Pantoea agglomerans* groups had minimum AICc for logistic. In contrast, all eight *Salmonella* Typhimurium comparisons and all eight *Oscillatoria* comparisons selected exponential. The group labelled *Tetraselmis tetrahele* had 11 Gompertz, nine logistic and five exponential minima among 25 curves.

Study-level patterns also differed. Within Bae, minima were 55 logistic, 31 Gompertz and two exponential among 88 curves. Within Gill, exponential had 21 of 25 minima.

The four examples showed different trajectory shapes (Figure 1). C041 increased and reached a plateau, which logistic and Gompertz followed closely. C034 increased and then declined below the fitted plateaus. C185 had no observed plateau, and all three fitted curves nearly overlapped. C246 varied little around its baseline. Both bounded fits for C185 had numerical warnings; for C246, only Gompertz was flagged.

3.3 Temperature patterns

Strict screening retained 41 of 88 Bae curves; relaxed screening retained 46. Retention differed across temperatures: 14/22 at 5 °C, 7/22 at 15 °C and 10/22 at both 25 and 35 °C. Eight strict series, containing 26 retained curves, had at least three temperatures.

All eight strict series had positive descriptive slopes of $\log r$ against temperature (Figure 3). Their highest-to-lowest retained temperature rate ratios ranged from 4.9 to 33.6, across different temperature spans. Seven series increased at every retained temperature; Enterobacter had a lower estimate at 25 than at 15 °C before increasing at 35 °C. Positive overall slopes therefore did not imply universal monotonicity.

The relaxed screen added two qualifying series. All ten relaxed series had positive slopes, and the eight shared series retained positive directions. Plateau estimates were less consistent (Figure 4). Five strict series increased at each retained temperature. *Chryseobacterium balustinum*, *Enterobacter* sp. and the group labelled *Stenotrophomonas maltophilia*.2 were nonmonotonic. For example, the *Enterobacter* plateau declined from 0.952 at 25 °C to 0.925 at 35 °C.

4 Discussion

I compared growth models across organisms and then examined temperature patterns within comparable series. Logistic was most often preferred, followed by Gompertz, and removing numerical warnings did not change this order. However, several models could have similar support for the same curve. As emphasised by Johnson and Omland (2004), model selection compares the proposed alternatives; it does not establish that the preferred model captures every process in the data.

The examples were consistent with the expectation that the observed growth phase affects model preference. Where growth and a plateau were both recorded, bounded curves described the trajectory more closely. Without a plateau, their capacity estimates could be poorly constrained even when the fitted lines almost coincided with exponential. Decline after growth caused a different problem: a simple sigmoid could not follow both the initial increase and the later

decrease. These patterns also limit the interpretation of differences among organism labels. The *Pantoea* and *Salmonella* groups came from different experimental settings, so their model preferences cannot be attributed to species biology alone.

AICc and BIC also gave different proportions of exponential minima. For a curve with six observations, the AICc correction is 12 for exponential ($k = 3$), but 40 for either bounded model ($k = 4$). This large difference can favour exponential in a short series. BIC has a different sample-size penalty. The disagreement therefore shows that conclusions depend partly on the comparison criterion, especially where sampling is sparse.

The positive rate slopes were consistent with faster bacterial growth at higher temperatures below the optimum, as described by Ratkowsky et al. (1982). However, the fitted logistic constant is a model parameter rather than a direct measurement of cell division. The *Enterobacter* series was also nonmonotonic, so a positive overall slope should not be read as a steady increase at every temperature. The range and number of temperatures were insufficient to locate thermal optima.

Plateau values did not show the same common direction. Faster reproduction need not produce a higher final cell number or biomass. In a phytoplankton experiment, Bernhardt et al. (2018) found that carrying capacity decreased with temperature and related this pattern to metabolism and body size. Its different organism and measurement scale limit direct comparison with the bacterial OD plateaus here. Optical density depends on light scattering and requires calibration before it can represent cell abundance quantitatively (Stevenson et al., 2016). The present K estimates therefore describe OD plateaus within each series, rather than measured cell-count carrying capacities.

Restricting temperature analysis to one study, medium and measurement scale reduced some confounding. Nevertheless, screening retained fewer curves at 15 °C than at the other temperatures. Selecting curves with clear growth and limited decline may exclude meaningful responses, including stress or mortality. The temperature patterns consequently apply to the retained series, rather than to all organisms in the compiled dataset.

Several limitations affect interpretation. Additive constant-variance errors may be inappropriate for measurements whose variability increases with magnitude. Repeated observations in a time series may also violate the independence assumption. Numerical flags depend on scaling and thresholds; their absence is not a confidence interval or a proof of identifiability. The current analysis does not quantify parameter uncertainty. A sensitivity screen addresses only some choices and cannot replace uncertainty estimation.

Measurement scale is a further limitation. The supplied CFU and N labels do not consistently identify transformations or normalisation. This is particularly relevant for nearly flat curves around a nonzero baseline. In addition, one source concerns experiments with temperature shifts (Zwietering et al., 1994), so a single supplied temperature label may not describe the complete experimental history. Keeping these curves in the broad comparison allowed their fitted shapes to be compared, but they were excluded from the restricted temperature analysis.

A likelihood model with variance increasing with the mean could give a different balance between early and late observations than least squares with constant variance. This would help distinguish a change in estimated growth from a change caused by the error assumption. A

Bayesian hierarchical model could estimate uncertainty in temperature effects while accounting for differences between strains and studies. Its joint distribution for rate and plateau would also show whether an apparent temperature trend was supported separately from uncertainty in K. Such analyses would need defensible measurement models and priors; they could not recover an unobserved plateau without additional assumptions. A predictive machine-learning model could provide a further benchmark, but validation should hold out complete curves or studies to avoid treating correlated observations as independent.

5 Conclusion

Logistic was the most frequently preferred model, followed by Gompertz, and this ranking persisted after removing numerical warnings. Preference varied across organism labels and studies. Differences in growth-phase coverage and sampling could contribute to this variation, but their effects were not tested separately. Within the screened Bae series, logistic growth rates generally increased with temperature, while plateau patterns were less uniform. Information criteria, numerical checks and observed curve shapes should therefore be considered together when interpreting these fits.

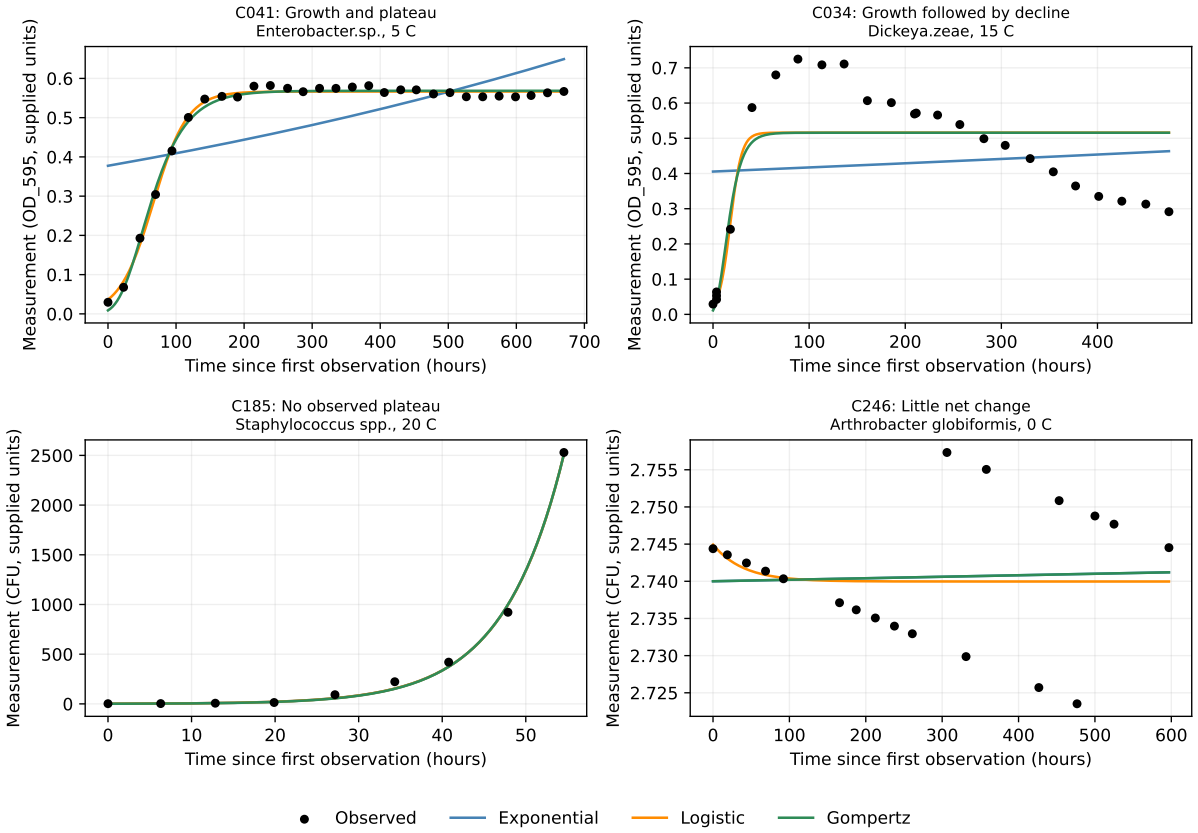


Figure 1: Illustrative observed trajectories and fitted growth models. C041 shows growth and a plateau, C034 growth followed by decline, C185 growth without an observed plateau, and C246 little net change. Examples were chosen to illustrate distinct limitations, not as a random sample. Lines evaluate fitted equations on a dense time grid. CFU labels are retained as supplied and do not confirm the original measurement transformation. Several C185 and C246 parameters have numerical warnings.

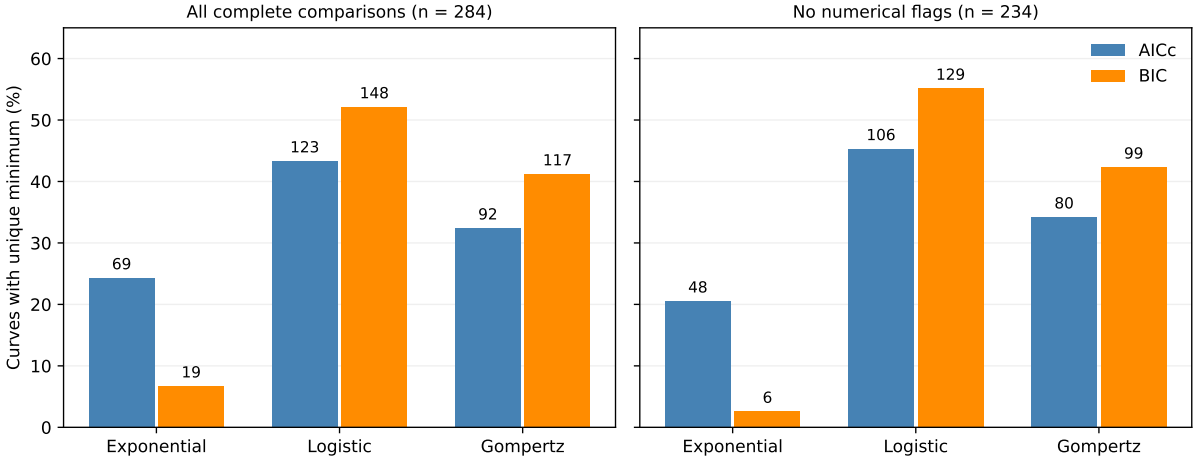


Figure 2: Relative model preference under AICc and BIC for 284 complete comparisons and the 234 comparisons without numerical warnings in any candidate model. Bar heights are percentages of curves with a unique minimum; numbers above bars are counts. Comparisons always use the same three candidate models within each retained curve. A preferred model may still have structured residuals or unreliable parameters.

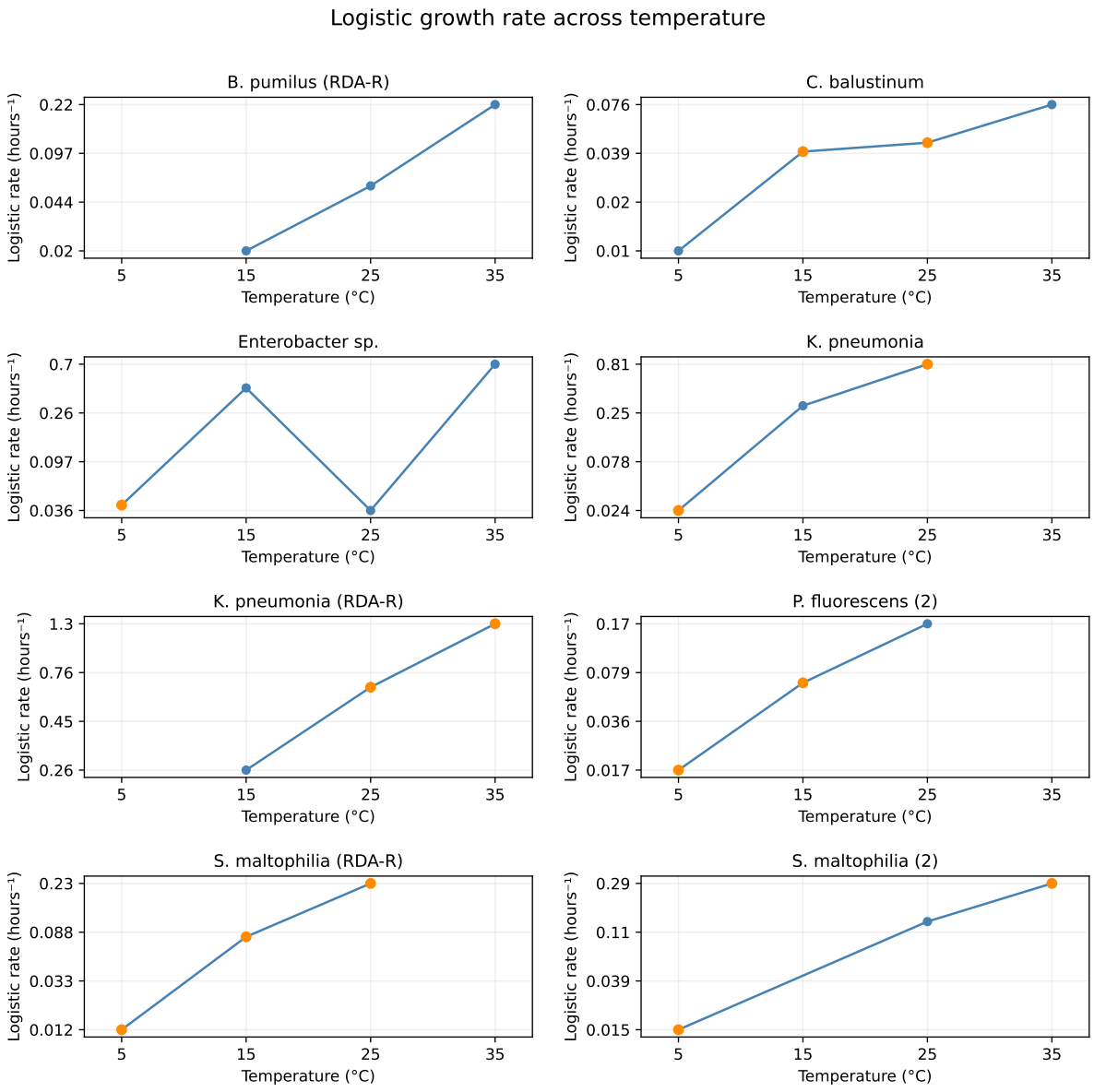


Figure 3: Logistic rate estimates in eight Bae-study series (TSB, OD595) with at least three temperatures after strict screening. The vertical axes are logarithmic and separately scaled. Orange points have logistic ΔAICc below two; blue points do not. Lines connect estimates and do not represent fitted temperature-response functions. Confidence intervals are not available. Panel names abbreviate the supplied labels: B., C., K., P. and S. denote Bacillus, Chryseobacterium, Klebsiella, Pseudomonas and Stenotrophomonas. Parentheses retain the supplied strain or group identifiers.

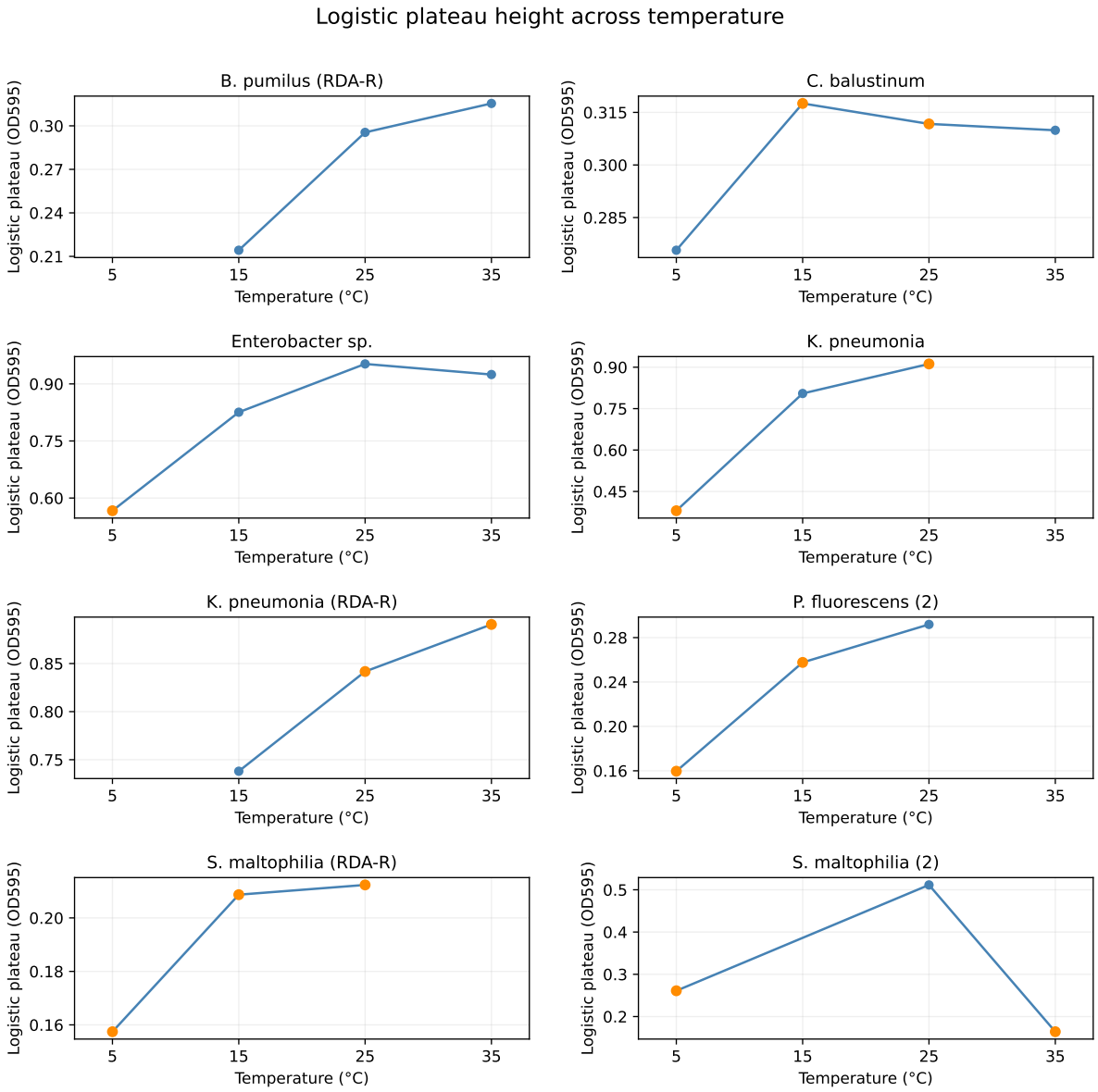


Figure 4: Logistic plateau estimates for the same eight screened series, with panel names and colours as in Figure 3. Each panel has a separate vertical scale. These are OD595 plateaus rather than cell counts. Five series increase at every retained temperature, while three have nonmonotonic patterns. Lines connect estimates; confidence intervals are not available.

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723.

Bae, Y. M., Zheng, L., Hyun, J. E., Jung, K. S., Heu, S., and Lee, S. Y. (2014). Growth characteristics and biofilm formation of various spoilage bacteria isolated from fresh produce. *Journal of Food Science*, 79(10):M2072–M2080.

Bernhardt, J. R., Sunday, J. M., and O'Connor, M. I. (2018). Metabolic theory and the temperature-size rule explain the temperature dependence of population carrying capacity. *The American Naturalist*, 192(6):687–697.

Galarz, L. A., Fonseca, G. G., and Prentice, C. (2016). Predicting bacterial growth in raw, salted, and cooked chicken breast fillets during storage. *Food Science and Technology International*, 22(6):461–474.

Gill, C. O. and DeLacy, K. M. (1991). Growth of Escherichia coli and Salmonella typhimurium on high-pH beef packed under vacuum or carbon dioxide. *International Journal of Food Microbiology*, 13(1):21–30.

Hurvich, C. M. and Tsai, C. L. (1989). Regression and time series model selection in small samples. *Biometrika*, 76(2):297–307.

Johnson, J. B. and Omland, K. S. (2004). Model selection in ecology and evolution. *Trends in Ecology and Evolution*, 19(2):101–108.

Phillips, J. D. and Griffiths, M. W. (1987). The relation between temperature and growth of bacteria in dairy products. *Food Microbiology*, 4(2):173–185.

Ratkowsky, D. A., Olley, J., McMeekin, T. A., and Ball, A. (1982). Relationship between temperature and growth rate of bacterial cultures. *Journal of Bacteriology*, 149(1):1–5.

Roth, N. G. and Wheaton, R. B. (1962). Continuity of psychrophilic and mesophilic growth characteristics in the genus Arthrobacter. *Journal of Bacteriology*, 83(3):551–555.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464.

Silva, A. P. R. D., Longhi, D. A., Dalcanton, F., and Aragão, G. M. F. D. (2018). Modelling the growth of lactic acid bacteria at different temperatures. *Brazilian Archives of Biology and Technology*, 61:e18160159.

Sivonen, K. (1990). Effects of light, temperature, nitrate, orthophosphate, and bacteria on growth of and hepatotoxin production by Oscillatoria agardhii strains. *Applied and Environmental Microbiology*, 56(9):2658–2666.

Stannard, C. J., Williams, A. P., and Gibbs, P. A. (1985). Temperature/growth relationships for psychrotrophic food-spoilage bacteria. *Food Microbiology*, 2(2):115–122.

Stevenson, K., McVey, A. F., Clark, I. B. N., Swain, P. S., and Pilizota, T. (2016). General calibration of microbial growth in microplate readers. *Scientific Reports*, 6:38828.

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: fundamental algorithms

for scientific computing in Python. *Nature Methods*, 17:261–272.

Zwietering, M. H., De Wit, J. C., Cuppers, H. G. A. M., and Van 't Riet, K. (1994). Modeling of bacterial growth with shifts in temperature. *Applied and Environmental Microbiology*, 60(1):204–213.

Zwietering, M. H., Jongenburger, I., Rombouts, F. M., and Van 't Riet, K. (1990). Modeling of the bacterial growth curve. *Applied and Environmental Microbiology*, 56(6):1875–1881.